Reg No.									

SATHYABAMA
INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)

CONTINUOUS ASSESSMENT EXAM-1 SET-A

Programme: BE(ECE,ECE with DS) Max. Marks: 50
Course Code: S13BLH22 Time: 2 Hours
Course Name: DIGITAL LOGIC CIRCUITS Sem: II
Batch : 2025-2029 Date:21-02-2026

Part-A Answer ALL the questions (7 × 2=14)

Q.No	Questions	CO
1.	Define number systems ?	CO1
2.	Perform binary addition for the given number $(1010)_2$ & $(0011)_2$	CO1
3.	Implement NOT gate using NAND gate	CO1
4.	Draw truth table and logic diagram for EX-OR gate?	CO1
5.	Define POS and SOP?	CO2
6.	Recall the steps to convert canonical POS to standard POS?	CO2
7.	Implement the combinational logic for the given function F=AB+BC	CO2

Part-B Answer ALL the questions (3 × 12 = 36)

8.	A. Compute the following Conversions (a) $(11101)_2 = (?)_{10}$ (b) $(1110101)_2 = (?)_{16}$ (c) $(24.32)_{10} = (?)_2$ (d) $(9)_2 - (6)_2 = (?)_2$ using 1's complement B. By using binary rule to divide the following binary number $(11011.1)_2$ by $(101)_2$	CO1
	(OR)	
9.	A. Explain with neat diagrams and truth table for the following gates (a) NAND Gate (b) NOR Gate (c) EX-OR Gate B. State and prove the demorgan's theorem	CO1

10.	Examine and simplify the following logic expressions (a) F=ABC+A'B+ABC' (b) F=C(B+C)(A+B+C) (c) F=(X+Y)'(X'+Y')	CO1
	(OR)	
11.	Apply K-map to reduce the following function. (a) F(A,B,C,D)=Σm(0,2,5,6,7,8,10,14) (b) F(A,B,C,D)=ΠM(1,3,5,7,13,15)	CO2

12.	Obtain the canonical SoP expression of the function (a) Y(A,B)=A+B (b) Y(A,B,C)=A+ABC (c) Y(A,B,C)= AC+AB+BC	CO2
	(OR)	
13.	Apply K-map to reduce the following function. (a) F(A,B,C,D)= ΠM(0,2,3,8,9,12,13,15)	CO2